INCH-POUND

MIL-S-24187/3B(SH)
22 March 1991

SUPERSEDING
MIL-S-24187/3A(SH)
9 September 1983

MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL OPERATION, STYLE CLS

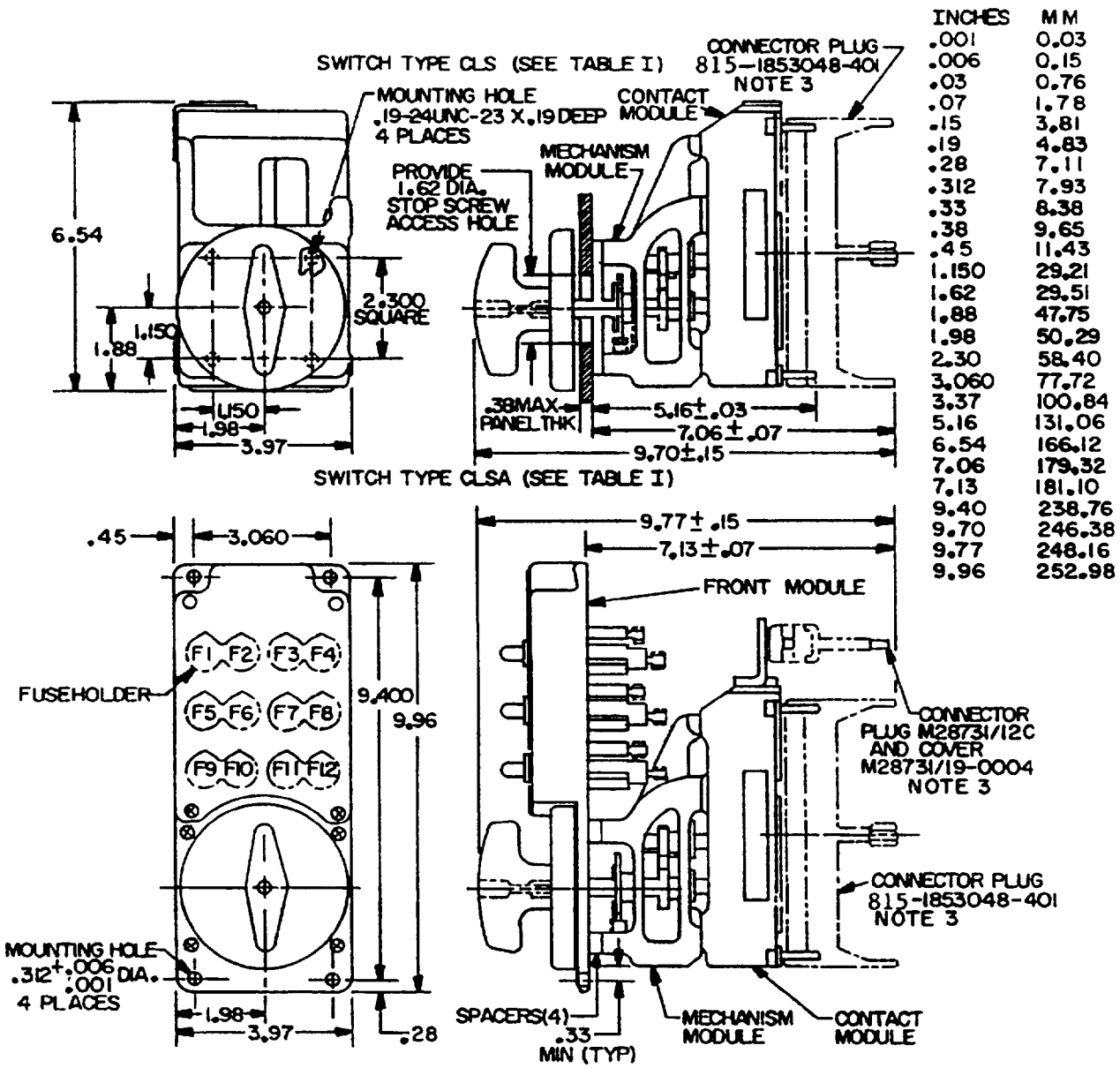
This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.

AMSC N/A FSC 5930

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INCHES	MM
.001	0.03
.006	0.15
.03	0.76
.07	1.78
.15	3.81
.19	4.83
.28	7.11
.312	7.93
.33	8.38
.38	9.65
.45	11.43
1.150	29.21
1.62	29.51
1.88	47.75
1.98	50.29
2.30	58.40
3.060	77.72
3.37	100.84
5.16	131.06
6.54	166.12
7.06	179.32
7.13	181.10
9.40	238.76
9.70	246.38
9.77	248.16
9.96	252.98

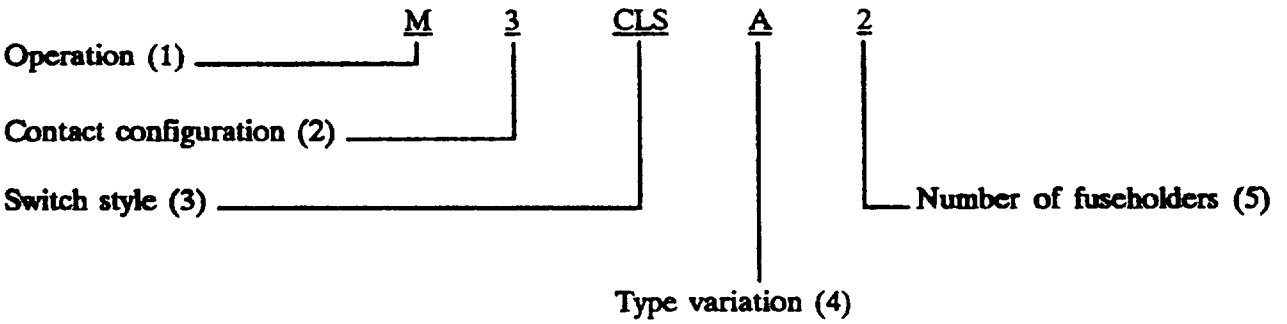
NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are plus or minus 0.02 (0.51 mm) for two-place decimals and plus or minus 0.005 (0.13 mm) for three-place decimals.
3. Supplied only when specified.
4. Metric equivalents (to the neareast 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm

FIGURE 1 *CLS switch assembly*

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Type designation:



1. Switch assemblies shall be manual operated designated M.
2. Contact configuration shall be type 2, 3, or 4, as specified (see figure 3).
3. Switch style shall be designated by the letters CLS.
4. Switch assemblies with no provisions for fuseholders shall be designated 0. Switch assemblies shall be provided with up to six double circuit fuseholders, and shall be designated A.
5. Number of additional double circuit fuseholders installed shall be indicated by a single digit, 0 to 6.

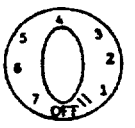
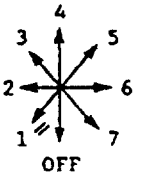
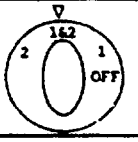
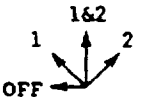
TABLE I. *Switch selection.*

Type designation				Application data					
Switch operation	Contact configu-ration	Switch style	Additional fuseholders	Switching action	Stop position[1]	Dial position[2]	Handle position[3]	Wired contacts[4]	Unwired contacts[4]
M M	2 2	CLS CLSA	0 0-6	1-2-OFF	2 & 7			( )D LOAD 2 ( )C INPUT ( )B LOAD 1	( )E
M M	2 2	CLS CLSA	0 0-6	1-OFF-2	2 & 7			( )E LOAD 2 ( )D INPUT/OFF ( )C ( )B LOAD 1	

See footnotes at end of table

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Switch operation	Contact configu-ration	Switch style	Additional fuseholders	Switching action	Stop position[1]	Dial position[2]	Handle position[3]	Wired contacts[4]	Unwired contacts[4]
M	3	CLS	0	1-4-7 -OFF	4	▽  _5	 _5	HH, HE INPUT/OFF GE,GD LOAD 7 FD,FC LOAD 6 EC,EB LOAD 5 DB,DE LOAD 4 CE,CD LOAD 3 BD,BC LOAD 2 AC,AB LOAD 1	XB, XX
M	3	CLSA	0-6						
M	4	CLS	0	OFF-1-1 &2-2	3&7	▽ 		( )E LOAD 2 ( )D INPUT ( )C LOAD 1 ( )B OFF	
M	4	CLSA	0-4						

Example: Type designation for a manual switch assembly, with type 2 contact configuration, and six additional fuseholders is M2CLSA6 The same switch assembly without fuseholders is M2CLSA0.

[1]Stops shall be adjustable to limit switch handle rotation to the positions indicated or to any consecutive positions within the maximum rotation indicated.

[2]Reference position of a CLS switch is considered to be that in which the flat on the handle shaft is up. The mark ▽ and the dial positions shown are used to designate the reference position.

[3]Normal position of a CLS switch is the position which coincides with normal circuit conditions in any application. The handle shall be adjustable to point upward in the normal position or to any specified position, in 45 degree increments from upward. The handle and rotary description plate rotate.

[4]See figures 2 and 3 for complete description of contact identification.

[5]Limit of rotation

REQUIREMENTS:

Construction: Style CLS switch parts, modules and assemblies shall be constructed in accordance with Drawing 803-4680150.

Characteristics: Characteristics shown in table II shall apply to the style CLS switch assembly.

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TABLE II. *Characteristics.*

Characteristics	Value		
	Contact configuration		
	2	3	4
Number of poles	56	24	56
Number of positions	2 or 3	2 to 8	4
Break-before-make	Yes	Yes	–[1]
Electrical rating and endurance loading:			
10 amperes, 115 V, 60 Hz, unity p.f.	20,000 Operations	15,000 Operations	20,000 Operations
7.5 amperes, 115 V, 60 Hz, unity p.f.	30,000 Operations	20,000 Operations	30,000 Operations
5 amperes, 50 V, dc, resistive	30,000 Operations	20,000 Operations	30,000 Operations
2 amperes, 115 V, dc, resistive	30,000 Operations	20,000 Operations	30,000 Operations
25 microamperes, 1.5 V, dc, resistance	30,000 Operations	20,000 Operations	30,000 Operations
Contact resistance:			
Switch contacts:			
At 10 amperes	16 milliohm		
At 2.5 microamperes	50 milliohm		
Torque – shaft rotation	40 pound-inches		
Stop strength	100 pound-inches		
Weight: CLS	5 pounds		
CLSA 6	10 pounds		
Temperature and humidity (for rotational torque and endurance tests)			
Minimum	Minus 20 °F (minus 29 °C)		
Maximum	Plus 149°F (plus 65°C) and 40 to 50 percent rh		
Shock	MIL-S-901, grade A, class 1, type C		
Vibration	MIL-STD-167-1, type I, except variable frequency		
Temperature rise	30 °C		

[1]Switch is a combination of make-before-break (MBB) and break-before-make (BBM). MBB is provided between 1 or 2 and 1 and 2 positions. All other functions are BBM (see figure 2).

Switch assembly: modules

Style CLS switch assembly shall consist of the following modules as detailed on Drawing 803-4680150:

Front module, CLSA only
Mechanism module
Switch contact module

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Switch function: Style CLS switch assemblies shall provide the switching functions as shown on figure 2. Contact configuration 2 shall provide 56 two- or three-position poles, each consisting of four stationary contacts, B, C, D, and E and a movable contact. Contact configuration 3 shall provide 24 two- to eight-position poles, each consisting of nine stationary contacts X, A, B, C, D, E, F, G, and H and a movable contact. Contact configuration 4 shall provide 56 four-position poles, each consisting of three stationary contacts C, D, and E and a movable contact. Stops shall prevent transfer of the movable contacts into unused positions. Stops (removable) shall be provided which, when inserted as specified in table I, will limit movement to consecutive used positions. Stops shall be as detailed on Drawing 803-4680150 and shall be accessible from the front of the switch assembly, after removal of the handle and rotary description plate.

Contact designation: Each stationary switching contact shall be identified by two letters and a number. The first letter shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figure 2. The number shall be that of the column in which the contact is located (see figure 3). Each contact in the switch contact module connector plug shall be assigned the same identification as the contact in the switch contact module to which it connects.

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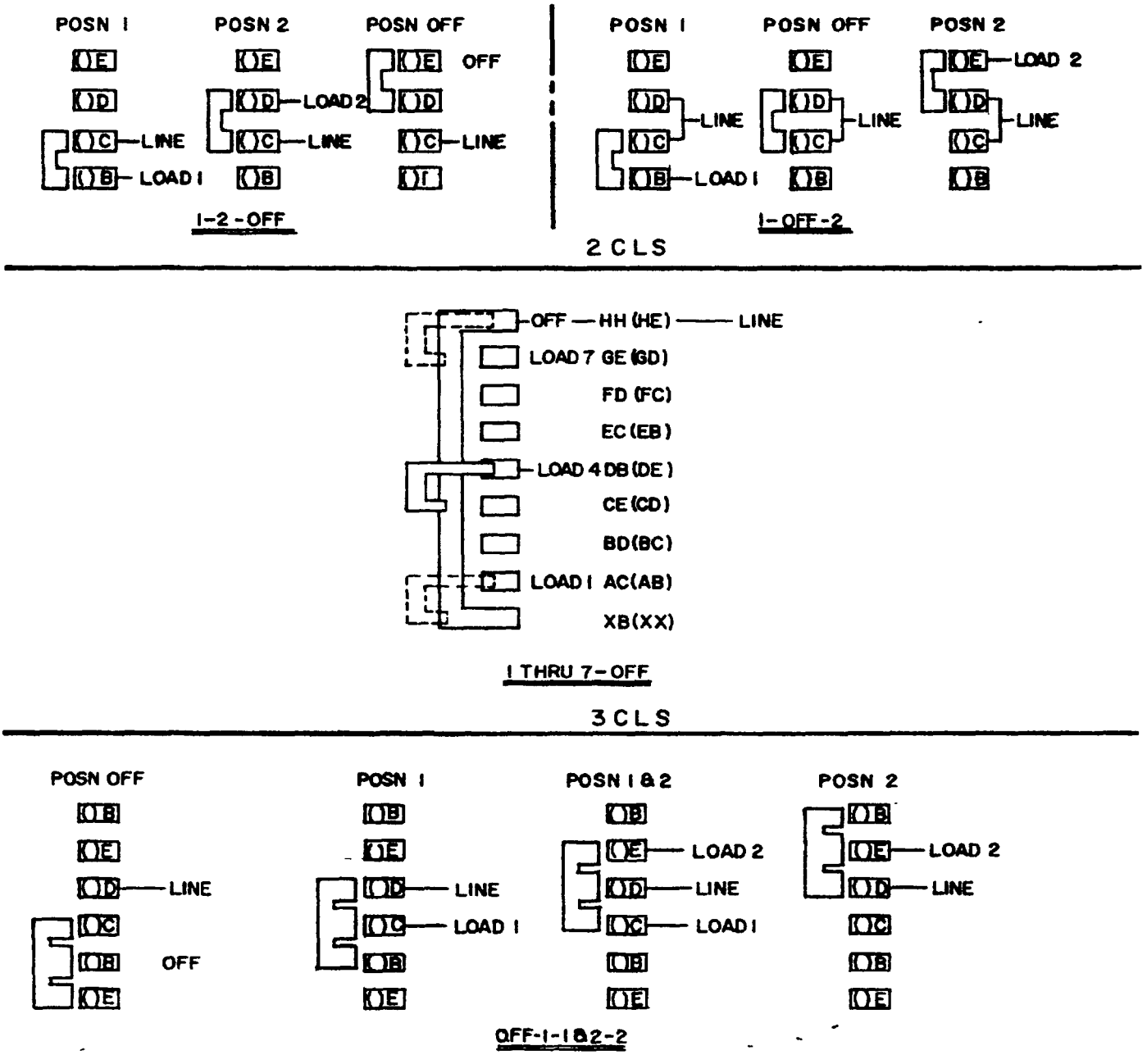
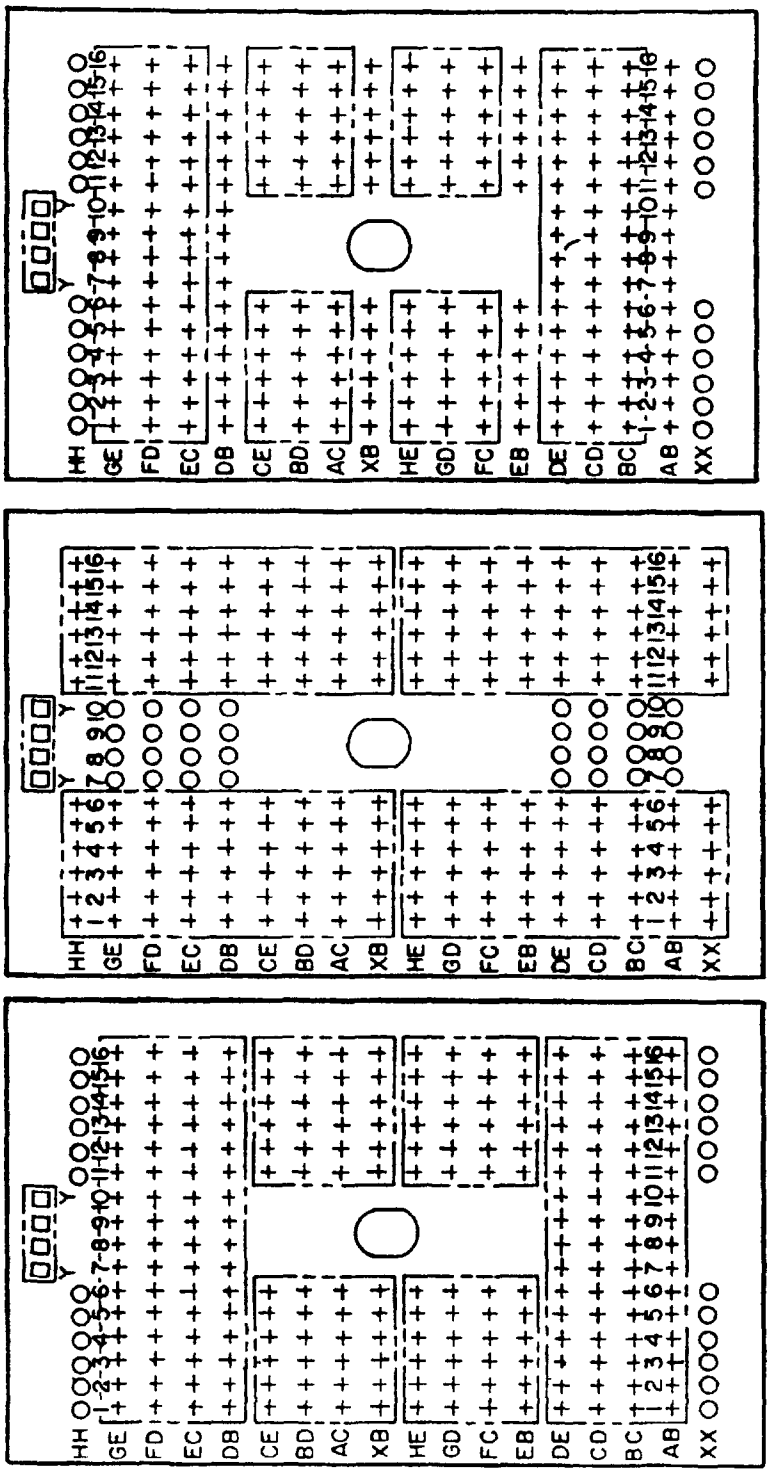


FIGURE 2. *Contact configuration.*

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NOTE: Bushings and contacts shall be installed in the stationary plate (815-1853048-301) as follows. Circles signify spaces without bushings and contacts. Crosses signify bushings and contacts which are installed. Squares, or the Y contacts are not used. Dashed rectangles enclose groups of poles, are shown for information only. Contact location markings shall be molded onto the front and rear surfaces of connector plugs (815-1853048-401).

FIGURE 3. *Stationary plate, rear view.*

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Fuseholder connector wiring, 38-pin receptacle M28731/11F(803-4680150-110): Wiring shall be as shown in table III.

TABLE III. *Fuse connections.*

Fuse number	Contact	
	Line	Load
1	D	K
2	C	J
3	B	F
4	A	E
5	P	U
6	N	T
7	M	S
8	L	R
9	W	Y
10	BB	HH
11	AA	DD
12	V	X
–[1]	PP	BFI return
–[2]	TT	BFI supply

[1]Fuseholder contact D shall be wired in parallel.
[2]Fuseholder contact C shall be wired in parallel.

Repair parts: Repair parts shall be specified in table VI.

TABLE IV. *Repair parts.*

Drawing 803-4680150 piece No.		Description	Quantities required for quantity of switches installed					
			1–3	4–12	13–30	31–60	61–120	121 or more
CLSA	CLS							
101	1	Switch, type M2CLS/2CLSA6	1	1	1	1	2	3
102	2	Switch, type M3CLS/3CLSA6	1	1	1	1	2	3
103	3	Switch, type M4CLS/4CLSA6	1	1	1	1	2	3
41	41	Detent spring	1	1	1	2	3	6
35	35	Stop screw	1	1	1	2	3	4
109		Fuseholder	1	2	3	4	5	6
		Fusecarrier (1A0981)	1	2	3	4	5	6
		Fuse (load circuit)	See note 1					

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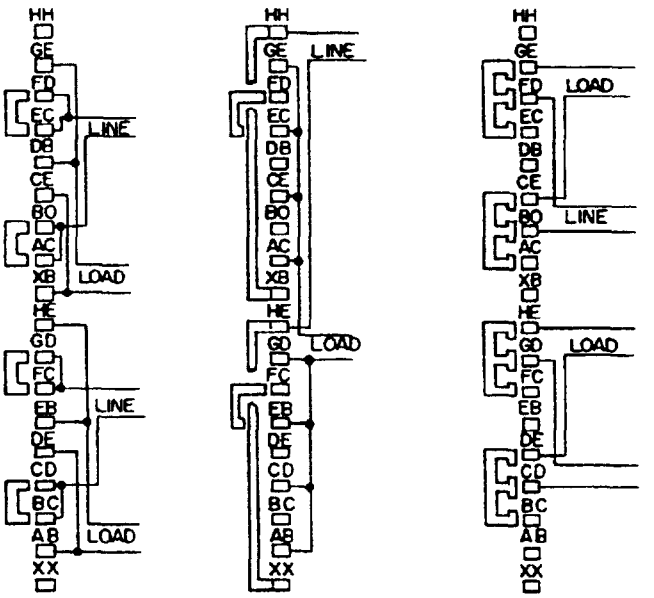
NOTES: 1. One hundred fifty percent repair fuses shall be supplied. Fuses shall be supplied in multiples of five, in standard commercial packages, with a minimum quantity of 15.

2. Connector plug, piece 8, is not to be supplied.

Qualification samples: Two switch assemblies of each contact configuration shall be submitted for qualification inspection.

TESTS:

Loading of switches: Loading of switches for endurance and temperature rise tests for qualification and quality conformance, group C testing, shall be in accordance with figure 4 and table V.



TYPE 2CLS			TYPE 3CLS			TYPE 4CLS		
COLUMNS	VOLTS	AMPS	COLUMNS	VOLTS	AMPS	COLUMNS	VOLTS	AMPS
1	115	10 AC	1 AND 2	115	10 AC	1	115	10 AC
5	50	5 DC	5 AND 6	50	5 DC	5	50	5 DC
7 AND 9	115	7.5 AC	11 AND 12	115	2 DC	11	115	2 DC
11	115	2 DC	16	1.5	25μ DC	16	1.5	25μ DC
16	1.5	25μ DC	13 AND 14	115	7.5 AC	7 AND 9	115	7.5 AC

FIGURE 4. *Test loading*

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Temperature rise: Contacts to be monitored during temperature rise test are specified in table V.

TABLE V. *Temperature measurements*

Column	Contacts		
	Type 2CLS	Type 3CLS	Type 4CLS
1	EC, DB	HH, GE	GE, FD
5	EC, DB	HH, GE	GE, FD

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 5930-N681-04)

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